

No. of Functional Features Included in the Solution

Date	03 July 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features implemented in the **Rising Waters – Intelligent Flood Prediction and Early Warning System**. The application uses a Decision Tree Machine Learning model to predict flood risk based on historical weather and rainfall data and provides users with an easy-to-use web interface for flood prediction.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Home Page	Displays project overview, features, and navigation	User Interface	Done	High
2	Weather Data Input	Allows users to enter weather and rainfall parameters	Prediction Module	Done	High
3	Input Validation	Validates user inputs before prediction	Backend Logic	Done	Medium
4	Flood Prediction	Predicts flood or no flood using the Decision Tree model	Machine Learning Model	Done	High
5	Prediction Result	Displays prediction result with safety recommendations	Result Module	Done	High
6	Error Handling	Handles invalid inputs and runtime exceptions	Backend Logic	Done	Medium
7	Responsive User Interface	Supports desktop and mobile devices	User Interface	Done	Medium
8	Model Integration	Loads the trained .pk1 model for prediction	ML Integration	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	Home Page, Weather Input, Flood Prediction, Prediction Result, Model Integration

Additional / Nice-to-Have Features	Responsive UI, Error Handling, Improved User Experience
Features Tested & Verified	8 / 8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Prediction Form, Result Page
2	Backend / Logic	3	Input Validation, Prediction Processing, Error Handling
3	Database / Storage	1	Historical Dataset, Trained Model (decision_tree_flood.pkl)
4	API / Integration	1	Machine Learning Model Integration (Future: Weather API)
5	Security / Authentication	1	Input Validation and Exception Handling